



HK IMO Training 2023-2024
Problem Set 10
For 16 December 2023



Problem 37. Let P be a point inside an equilateral triangle with height 1. Let x, y, z be the distances from P to the three sides of the triangle. Prove that

$$x^2 + y^2 + z^2 \geq x^3 + y^3 + z^3 + 6xyz.$$

Problem 38. Let x and y be rational numbers such that both $x + y$ and $x^2 + y^2$ are integers. Prove that x and y are integers.

Problem 39. There are 100 prisoners. Each prisoner is given a hat coloured in one of five colours. Everyone can see all hats except his own. Afterwards, the guard asks the prisoners one by one whether he knows the colour of his hat or not. If the prisoner answers 'no', he will be put in the jail. If the prisoner answers 'yes', he needs to write down the colour that he guesses. If this colour agrees with the colour of his hat, he will be freed, otherwise he will be put in the jail. Suppose all other prisoners can hear the answer of the prisoner but cannot see the colour he writes down, and they can know the outcome immediately. If the prisoners can think of a strategy together before they are given the hats, but cannot communicate afterwards, what is the maximum number of prisoners who are guaranteed to be freed?

Problem 40. Let $\triangle ABC$ be an acute triangle with $AB \neq AC$. Let D, E, F be the midpoints of BC, CA, AB respectively. Let M and N be the midpoints of the arc BC of the circumcircle of $\triangle ABC$ such that M and A lie on different sides of BC , while N and A lie on the same side of BC . Let I, J_1, J_2, J_3 be the incentre, D -excentre, E -excentre and F -excentre of $\triangle DEF$ respectively. Prove that the circumcircles of $\triangle ABC$, $\triangle NIJ_1$ and $\triangle MJ_2J_3$ meet at a common point.